Connecting Mathematics and Programming

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Abstract

To the author, mathematics is a feasible answer for the value of humanity in the era of emerging AIs. While machines ease our workload from heavy computation and routine reasoning, we can afford more time to focus on the purity and beauty of mathematics and acquire an appreciation. With this important appreciation, one can cultivate a robust mind capable of enduring even adversities in one's life journey and equip oneself for better control over powerful tools like programming.

In attempts to connect the mathematics of pure beauty and the programming for powerful computation, the author collaborated with various parties on Python workshops for counting, basic statistics, deep neural networks, etc. More recently, the author utilized generative AIs like DeepSeek to continue the endeavor in this direction. As a result, here is the convenient transformation of mathematical ideas into associated coding for fast development of learning materials when the coding process is accelerated significantly with generative AIs.

Introduction

Following past common practices, as seen in articles [1] and [2], the relevant source code for this work is made available in a GitHub repository under the MIT license open-source terms. The address this time is https://github.com/EssenceBL/MakingFriends. The content includes new interactive Python notebooks on several topics:

1. Study notes on touching circles with integral curvatures
2. Use of polynomials in information retrieval
3. Visualization of some parametrized curves

The first topic contains elementary reflections on attending the Shaw Prize Lecture 2024 on integral Apollonian packing [3]. The second and third topics serve as extra ingredients that were originally developed to demonstrate the interplay between mathematics and computational thinking to a group of summer interns at HKU.

Most modern browsers can run these Python notebooks with a simple login to the Colab environment. Such an environment also allows one to incorporate granular toolkits for holistic development. In particular, interactive plots made with GeoGebra are embedded naturally. In fact, all figures presented here are captured from the interactive assets in these shared notebooks.

Accordingly, this article just adopts materials I wrote in these notebooks and will be organized as a series of activities. While some activities are like theorems and sketches of proofs, they are still called activities. This is because they are associated to interactive materials in shared notebooks, that would encourage discussion and relaxation in the teaching and learning process.

Touching Circles on a Straight Line: Let's Greet the Year 2025

Activity 1. Let L be a common tangent line of three pairwise mutually tangent circles C_1, C_2, C_3. Let b_1, b_2 and b_3 be the respective curvatures (also known as bends, and may be thought of as reciprocals of radii, with a sign determined by viewing perspective) of these three distinct circles. Check the following symmetric relation with the help of some geometric construction:

$$(\mathrm{b}_1 + \mathrm{b}_2 + \mathrm{b}_3)^2 = 2(\mathrm{b}_1^2 + \mathrm{b}_2^2 + \mathrm{b}_3^2).$$

Deduce that, if both b_1 and b_2 are perfect squares, then so is b_3.

Check and Discussion 1. Denote centers of C_1, C_2, and C_3 by O_1, O_2, and O_3, and retrieve their radii as reciprocals: $\mathrm{r}_1 = \frac{1}{\mathrm{b}_1}$, $\mathrm{r}_2 = \frac{1}{\mathrm{b}_2}$, and $\mathrm{r}_3 = \frac{1}{\mathrm{b}_3}$. Without loss of generality, assume L is a horizontal line and C_3 is the smallest among the three circles. One could then follow the geometric construction as captured in Figure 1. It is now easy to get the horizontal distance between O_1 and O_2 by Pythagoras' Theorem:

$$\sqrt{(\mathrm{r}_1+\mathrm{r}_2)^2 - (\mathrm{r}_1-\mathrm{r}_2)^2} = 2\sqrt{\mathrm{r}_1\mathrm{r}_2}.$$

Similarly, the horizontal distance between O_1 and O_3 is $2\sqrt{\mathrm{r}_1\mathrm{r}_3}$; and the horizontal distance between O_2 and O_3 is $2\sqrt{\mathrm{r}_2\mathrm{r}_3}$. And now, one should be ready to formulate:

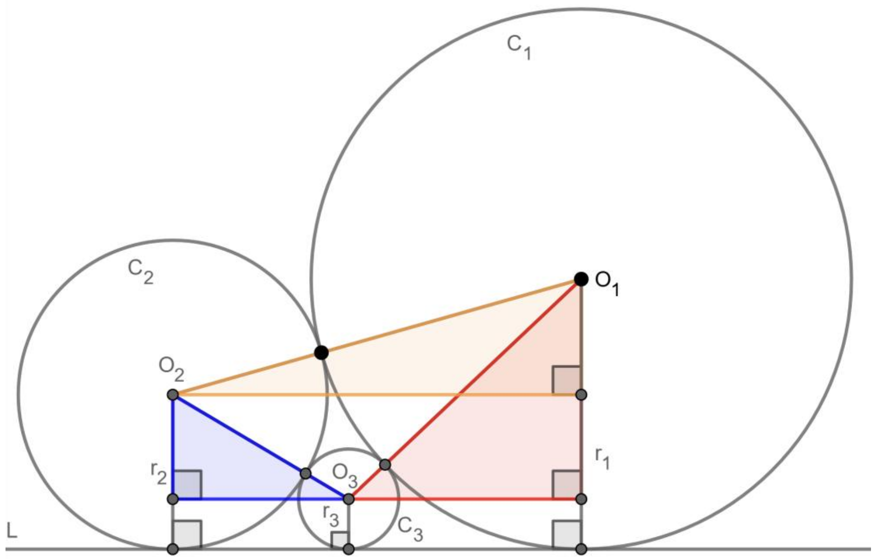

Figure 1

$$
\begin{array}{lrcl}
 & 2\sqrt{r_1 r_2} & = & 2\sqrt{r_1 r_3} + 2\sqrt{r_2 r_3} \\
\Leftrightarrow & \frac{1}{\sqrt{r_3}} & = & \frac{1}{\sqrt{r_2}} + \frac{1}{\sqrt{r_1}} \\
\Leftrightarrow & \sqrt{b_3} & = & \sqrt{b_2} + \sqrt{b_1} \\
\Rightarrow & b_3 & = & b_1 + b_2 + 2\sqrt{b_1 b_2} \\
\Rightarrow & b_3 - b_1 - b_2 & = & 2\sqrt{b_1 b_2} \\
\Rightarrow & b_1^2 + b_2^2 + b_3^2 + 2b_1b_2 - 2b_1b_3 - 2b_2b_3 & = & 4b_1b_2 \\
\Rightarrow & b_1^2 + b_2^2 + b_3^2 & = & 2b_1b_2 + 2b_1b_3 + 2b_2b_3 \\
\Rightarrow & (b_1 + b_2 + b_3)^2 & = & 2(b_1^2 + b_2^2 + b_3^2).
\end{array}
$$

The symmetric relation is verified. Conversely, if one solves for b_3 in b_1 and b_2, there would be two solutions, namely, $(\sqrt{b_1} \pm \sqrt{b_2})^2$. From this expression it's clear that, if both b_1 and b_2 are perfect squares, then so is b_3. Note that two solutions exist: if one first starts with two mutually tangent circles and a common tangent line, then there are generally two ways to construct a third circle tangent to these three objects. The curvature for one solution is $(\sqrt{b_1} + \sqrt{b_2})^2$ and for the other is $(\sqrt{b_1} - \sqrt{b_2})^2$. In generalization, we include the case $b_1 = b_2$, where the third "circle" could be a circle with curvature $4b_1$ or a *generalized circle* with curvature 0, which means a straight line!

Activity 2. The author pairs with DeepSeek to write code that drawn from an initial configuration of two mutually tangent circles with curvature 1 touching a straight line; and drawn more circles in iterations, such that each circle is tangential to the straight line, and two other circles in some previous iterations. Readers may find the code in shared notebooks, run them and appreciate the beauty of mathematical objects.

Check and Discussion 2. It is confirmed in Figure 2 that the resultant code run to produce the desired plot. The initial line is the horizontal line that all circles touch. The initial circles, with a curvature of 1, are only partially shown, while the curvatures of the other circles are displayed around their centers.

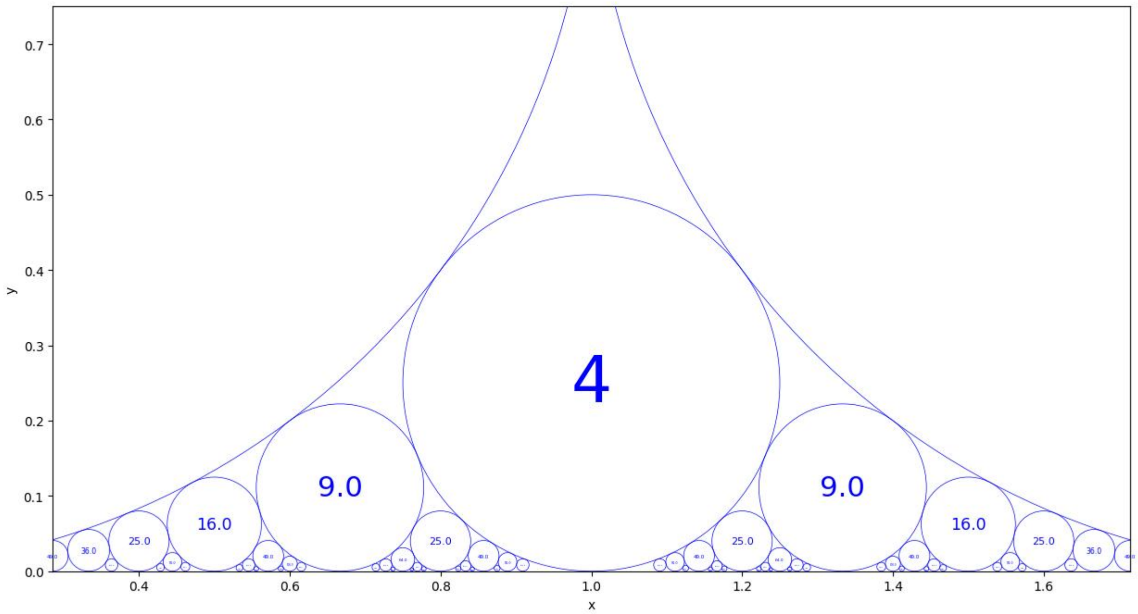

Figure 2

In further discussion, the plot is part of Ford Circles, dilated by a factor of two. Often, in-depth mathematical knowledge is beneficial for developing efficient programming. Keeping Ford Circles in mind, the author knew that positions of these circles are closely related to the idea of the Stern–Brocot tree, that generates rational numbers. This provides a good instructive strategy for AI to assist in practical programming. The coding experience with AI is comparable to working with past collaborators, promoting continuous learning for all participants.

There were also times when some strategies exceed the expectation of the AI model used; as usual, iterative coding is beneficial for rectifying misinterpretations in pair programming. It should also be understood that it often requires more knowledge to produce work than to comprehend the product. Therefore, the additional terms here should serve only as options for the audience to explore in their own areas of interest, whether in mathematical theory or coding exercises. In either way, or both, I wish them a rich journey.

Activity 3. Note that 2025 is a perfect square. Could we find a circle with curvature 2025 in the iterative process described in Activity 2?

Check and Discussion 3. The answer is yes. In fact, the code in Activity 2 allows one to trace new circles generated. Figure 3 captures a circle with curvature $2025 = 45^2$, which is tangential to the two larger circles with curvatures $28^2$and 17^2. And the code continues to generate more smaller circles with larger curvatures in iterations as shown in the figure. A further exercise for readers is to find all circles with curvature 2025, with iterations begin from two circles of curvature 1. Should there be 24 of them in total? We leave it to the readers.

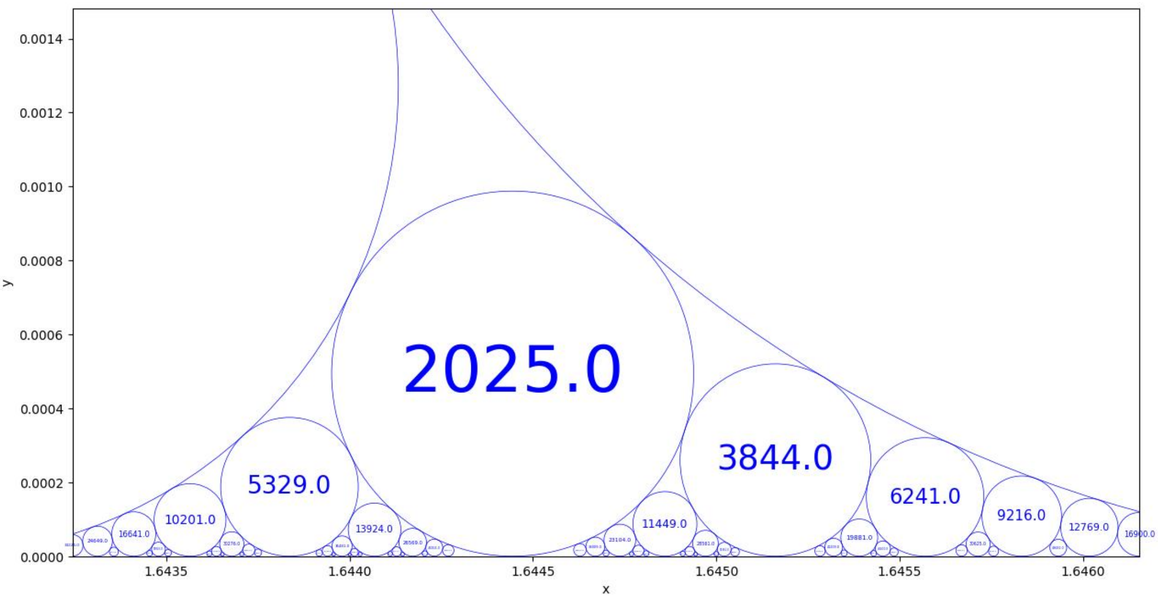

Figure 3

From the Problem of Apollonius to the Descartes Circle Theorem: An Approach Through Inversion

Now, suppose one is given three pairwise mutually tangent circles without a common line tangent to all of them. Among other setups, Apollonius of Perga studied a fourth circle touching (in other words, tangent to) all of them. Indeed, for this initial setup, there are two choices. The **Descartes' Circle Theorem** asserts that for any of these choices, the curvatures of the four pairwise mutually tangent circles satisfy a special symmetric relation. The following five consecutive activities in this section sketch a proof by using *inversion*.

Activity 4. An *inversion* is a transformation in the plane with reference to a circle C with center O and radius r. The inversion maps any point P distinct from O to a point P' on the ray OP, such that OP and OP' are in the same direction and

$$|OP||OP'| = r^2.$$

To extend the map for completeness, one typically maps the center O to a "point at infinity", and vice versa. Figure 4 captures an interactive plot of this map. Check that inversion is involutory: i.e. suppose the inversion in C sends P to P' and P' to P'', then $P = P''$.

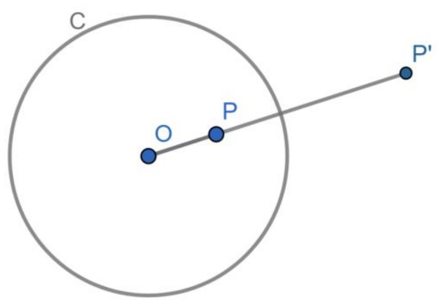

Figure 4

Check and Discussion 4. The checking is straightforward. One could further notice that the inversion transformation fixes points on the circle C. And by dragging the point P in the dynamic interactive plot, one may observe some additional properties of this transformation, such as:

(i) Inversion turns interior points of C to the outside and inverts exterior points of C to the inside.

(ii) The inversion induce transformation on geometric objects as a set of points. Inversion might deform shape of objects, but incidence is preserved.

Activity 5. Given a fixed reference circle C with radius r and center O. Let C_1 be a circle with radius ρ and its center distance d from O. Let C'_1 be the inversion of C_1 in C. In case $d > r$, check that C'_1 is also a circle.

Check and Discussion 5. One can construct two tangent lines from O to C_1 touching at points P_1 and P_2 respectively. Now, let A be any other point of C_1, the line ray OA will cut the circle C_1 at one more point, let's say B. Take inversion in C, these points on C_1 are mapped to P_1', P_2', A' and B' correspondingly as in Figure 5.

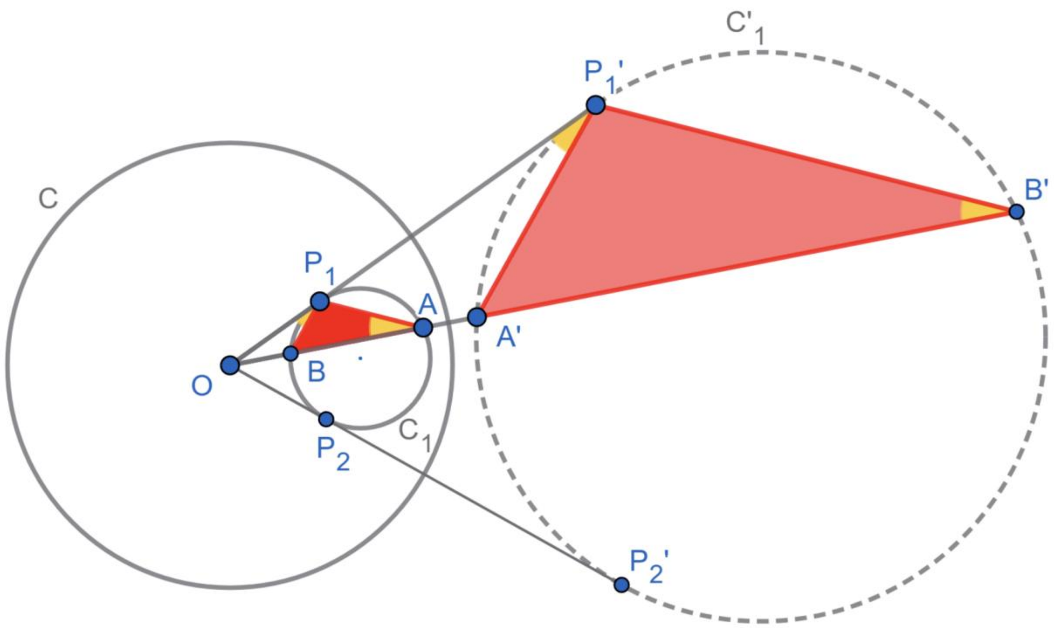

Figure 5

One could check by the definition of inversion that $\Delta OP_1A \sim \Delta OA'P_1'$ and $\Delta OP_1B \sim \Delta OB'P_1'$. Check by properties of similar triangles and tangent line that $\angle P_1'B'O = \angle BP_1O = \angle P_1AO = \angle A'P_1'O$. This implies that $P_1'B' \parallel P_1A$ and $A'P_1' \parallel BP_1$. In other words, with the role of A and B swapped, $\Delta P_1'A'B'$ is a dilation of ΔPBA from the center O with a fixed ratio $\frac{|OP'_1|}{|OP_1|}$. Therefore, inversion in C, the circle C_1 is mapped to another inverted circle C'_1 whose radius ρ' can be formulated as follows:

$$\rho' = \frac{\rho|OP'_1|}{|OP_1|} = \frac{\rho|OP'_1||OP_1|}{|OP_1|^2} = \frac{\rho r^2}{d^2 - \rho^2}.$$

Although a circle is mapped to a circle in inversion, it is also important to keep in mind that the center of a circle is often not the center after inversion.

Activity 6. Given a fixed reference circle C with radius r and center O. Let C_1 be a circle with radius ρ and its center distance d from O. Let C'_1 be the inversion of C_1 in C. In case $d < r$, check that C'_1 is also a circle.

Check and Discussion 6. One can construct a line joining the center of C and C'. Let such a line meet C_1 at P_1 and P_2, so that P_1P_2 is a diameter of C_1 and any other point A on C_1 satisfies $\angle P_1AP_2 = 90°$. The inversion in C maps P_1, P_2 and A to P_1', P_2' and A' respectively as in Figure 6.

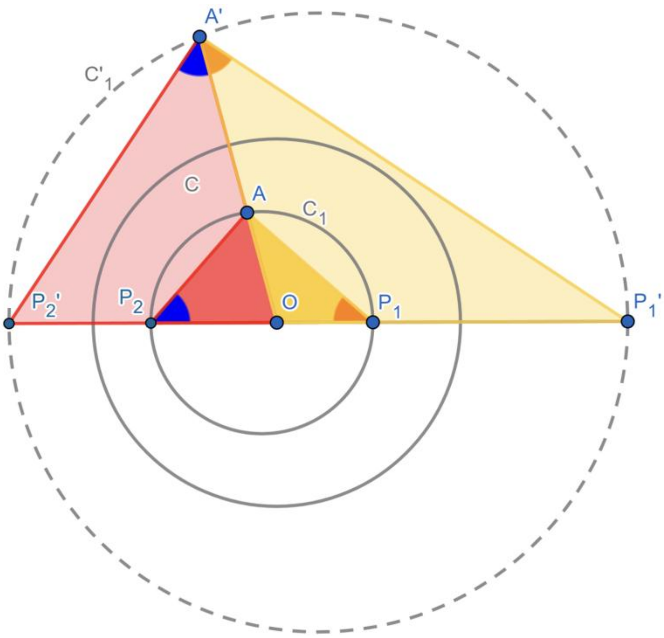

Figure 6

One could check by the definition of inversion that $\Delta OP_1A \sim \Delta OA'P_1'$ and $\Delta OP_2A \sim \Delta OA'P_2'$. Compute $\angle P_1'A'P_2' = 180° - \angle AP_1P_2 - \angle P_1P_2A = 90°$. Therefore, A' lies on a circle with a diameter $P_1'P_2'$. Moreover,

$$|P_1'P_2'| = |P_1'O| + |OP_2'| = \frac{r^2}{|P_1O|} + \frac{r^2}{|OP_2|} = \frac{r^2}{\rho - d} + \frac{r^2}{\rho + d} = \frac{2\rho r^2}{\rho^2 - d^2}.$$

Thus, taking inversion in C, the circle C_1 with radius ρ is mapped to a circle C_1' with radius

$$\rho' = -\frac{\rho r^2}{d^2 - \rho^2}.$$

Sometimes, the minus sign could make cancellation of an assigned sign for the curvature of a circle when viewing from interior versus exterior.

Activity 7. Given a fixed reference circle C with radius r and center O. Let C_1 be a circle with radius ρ and its center distance d from O. Let L'_1 be the inversion of C_1 in C. In case $d = r$, check that L'_1 is a straight line.

Check and Discussion 7. One can construct a point P on C_1 such that OP is a diameter of C_1. Let A be any other point of C_1. Let P' and A' be inversion points of P and A in C as in Figure 7.

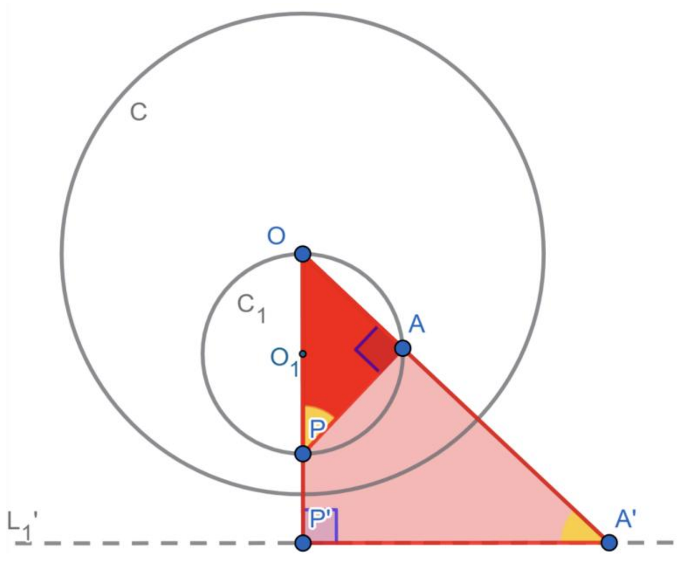

Figure 7

Check by the definition of inversion that $\Delta OAP \sim \Delta OP'A'$. Since OP is a diameter of C', $\angle OAP$ is a right angle. So $\angle OP'A'$ is also a right angle by corresponding angles of similar triangles. This means that A' lies on a line perpendicular to OP' at P'. Call this line L_1'. One can also see that any point on L_1' is the inversion in C of a certain point on C_1 by drawing a line joining this point back to O and see how it cut C_1.

Suppose further the line L_1' is at a distance $d' = |OP'|$ from O, one could formulate a relation between ρ and d' by

$$|OP||OP'| = r^2 \Rightarrow \rho = \frac{r^2}{2d'}.$$

Since inversion is involutory, a line not through O would be mapped to a circle through O in inversion in C. For a line through O, it will just be mapped to itself in inversion in C. To this end, through these activities, one checked that generalized circles are mapped to generalized circles in inversion and would be able do some calculation.

Activity 8. Descartes' Circle Theorem. Let C_1, C_2, C_3 and C_4 be four pairwise mutually tangent circles at six disjoint points. Let b_1, b_2, b_3 and b_4 be their respective curvatures, check that the following symmetric relation holds:

$$(b_1 + b_2 + b_3 + b_4)^2 = 2(b_1^2 + b_2^2 + b_3^2 + b_4^2).$$

Check and Discussion 8. Let's begin with three circles C_1, C_2 and C_3, mutually tangent at disjoint points $T_{i,j}$ between C_i and C_j for $1 \leq i < j \leq 3$. Suppose C_4 is a forth circle simultaneously tangent to all these three circles, the plan is to try construct C_4 as follows:

Let C be a circle with center $T_{1,2}$ and radius r. For convenience of later visualization and computation, one may choose r so that it is larger than diameters of both C_1 and C_2. Take inversion in C, C_1 is mapped to a straight line L_1'; C_2 is mapped to a straight line L_2'; C_3 is mapped to a circle C_3'. Since inversion preserves incidence and that $T_{1,2}$, the only incidence point of C_1 and C_2, is mapped to the point of infinity; L_1' and L_2' must be a pair of parallel lines. Moreover, $C_3{}'$ must be a circle tangent to this pair of parallel lines.

Now, one can choose a Cartesian coordinate system, such that the origin is the center of $C_3{}'$, the unit is the radius of $C_3{}'$, the direction of the parallel lines is the x-direction, the equation of $L_1{}'$ is $y = 1$ and that of $L_2{}'$ is $y =- 1$. To find circle simultaneously tangent to $L_1{}'$, $L_2{}'$ and $C_3{}'$, one can easily spot that there are exactly two:

(i) $C_{4+}{}'$ which is a circle centered at $(2, 0)$ with radius 1 unit; and

(ii) $C_{4-}{}'$ which is a circle centered at $(-2, 0)$ with radius 1 unit.

These circles in the required configuration are easier to spot after inversion. Moreover, inversion is involutory, now one can simply use inversion in C again, mapping $C_{4+}{}'$ to C_{4+} and mapping $C_{4-}{}'$ to C_{4-}. Since inversion preserves incidence, both C_{4+} and C_{4-} are tangential to C_1, C_2 and C_3 simultaneously as shown in Figure 8.

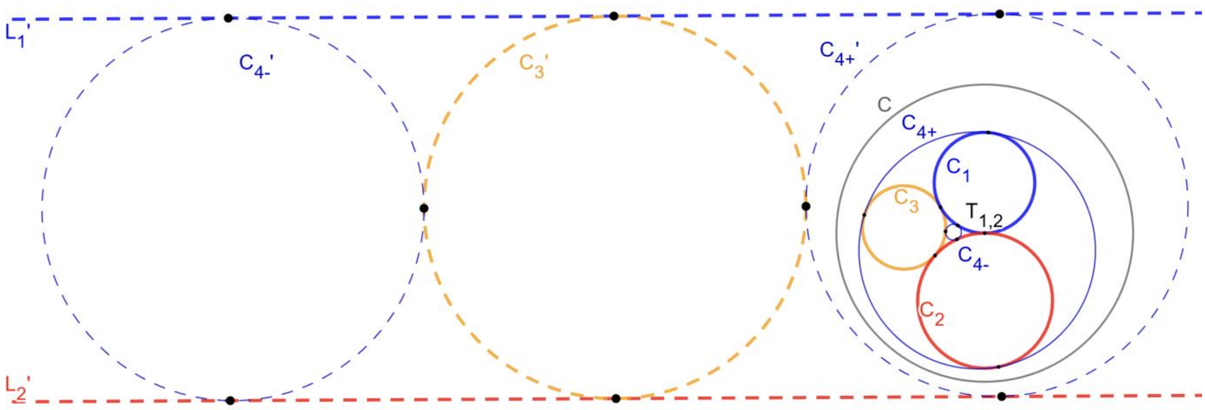

Figure 8

With a good choice of inversion, the construction of the fourth touching circle is made easy. One can continue to verify the symmetric relation on curvatures in the Descartes' Circle Theorem. In calculation, note that the choice of coordinates may change the metrics if dilation is involved, while rotation, translation and reflection won't affect it.

Denote by b_1, b_2, b_3, b_{4+} and b_{4-} the curvatures of C_1, C_2, C_3, C_{4+}

and C_{4-} respectively. Let coordinates of $T_{1,2}$ be (x, y), r be the radius and k be the factor of dilation in choice of coordinates. Note that the sign of curvature for a circle may become negative when the perspective changes from interior to exterior, and vice versa. Consider a negative sign when necessary, one can formulate the curvatures as the reciprocals of the radii using the formula discussed in previous activities:

$$\frac{k}{b_1} = \frac{r^2}{2(1-y)} \implies b_1 = \frac{2(1-y)k}{r^2},$$
$$\frac{k}{b_2} = \frac{r^2}{2(1+y)} \implies b_2 = \frac{2(1+y)k}{r^2},$$
$$\frac{k}{b_3} = \frac{r^2}{((x-0)^2+(y-0)^2)-1} \implies b_3 = \frac{(x^2+y^2-1)k}{r^2},$$
$$\frac{k}{b_{4\pm}} = \frac{r^2}{((x\mp 2)^2+(y-0)^2)-1} \implies b_{4\pm} = \frac{(x^2+y^2\mp 4x+3)k}{r^2}.$$

One is now ready to check:

$$b_1+b_2+b_3+b_{4\pm} = \frac{2k}{r^2}(x^2+y^2\mp 2x+3),$$
$$(b_1+b_2+b_3+b_{4\pm})^2 = \frac{4k^2}{r^4}(x^2+y^2\mp 2x+3)^2 \text{ and}$$

$$\begin{aligned}
&b_1^2+b_2^2+b_3^2+b_{4\pm}^2 \\
=&\frac{k^2}{r^4}[(2-2y)^2+(2+2y)^2+(x^2+y^2-1)^2+(x^2+y^2\mp 4x+3)^2] \\
=&\frac{2k^2}{r^4}[2^2+(2y)^2+(x^2+y^2\mp 2x+1)^2+(\pm 2x-2)^2] \\
=&\frac{2k^2}{r^4}[4+4y^2+(x^2+y^2\mp 2x+1)^2+4x^2\mp 8x+4] \\
=&\frac{2k^2}{r^4}[(x^2+y^2\mp 2x+1)^2+4(x^2+y^2\mp 2x+1)+4] \\
=&\frac{2k^2}{r^4}[(x^2+y^2\mp 2x+1)+2]^2 \\
=&\frac{2k^2}{r^4}(x^2+y^2\mp 2x+3)^2.
\end{aligned}$$

Therefore, one shall get the symmetric relation on b_1, b_2, b_3 and b_4 for both $b_4 = b_{4+}$ and $b_4 = b_{4-}$ as desired:

$$(b_1 + b_2 + b_3 + b_4)^2 = 2(b_1^2 + b_2^2 + b_3^2 + b_4^2).$$

It is also useful to rewrite the relation as a quadratic equation in b_4:

$$b_4^2 - 2(b_1 + b_2 + b_3)b_4 + 2(b_1^2 + b_2^2 + b_3^2) - (b_1 + b_2 + b_3)^2 = 0.$$

Consider the sum of roots in this equation, there is a simple relation between b_{4+} and b_{4-}:

$$b_{4+} + b_{4-} = 2(b_1 + b_2 + b_3).$$

Integral Apollonian Packing: A First Look

Activity 9. Given four generalized circles C_a, C_b, C_c and C_d pairwise mutually tangent at six disjoint points. Check that one could pick any one of them, and find a companion circle simultaneously tangent to the 3 other objects.

Check and Discussion 9. Following Activity 8, each of C_a, C_b, C_c and C_d has a companion denoted by $C_{a'}$, $C_{b'}$, $C_{c'}$ and $C_{d'}$ respectively, with desired tangent properties as drawn in Figure 9.

Let's look a bit deeper. Denote by (b_a, b_b, b_c, b_d) the 4-tuple (quadruple) of respective curvatures of the initial four touching circles C_a, C_b, C_c and C_d. These four companions suggest four different actions S_a, S_b, S_c and S_d that could transform this initial 4-tuple into another 4-tuple of curvatures of four touching circles:

$$S_a : (b_a, b_b, b_c, b_d) \mapsto (b_{a'}, b_b, b_c, b_d),$$
$$S_b : (b_a, b_b, b_3, b_d) \mapsto (b_a, b_{b'}, b_c, b_d),$$
$$S_c : (b_a, b_b, b_c, b_d) \mapsto (b_a, b_b, b_{c'}, b_d),$$
$$S_d : (b_a, b_b, b_c, b_d) \mapsto (b_a, b_b, b_c, b_{d'});$$

where $b_{a'}$ is the curvature of $C_{a'}$ in the first action; and following Activity 8, one may compute $b_{a'} =- b_a + 2(b_b + b_c + b_d)$; and likewise for other actions.

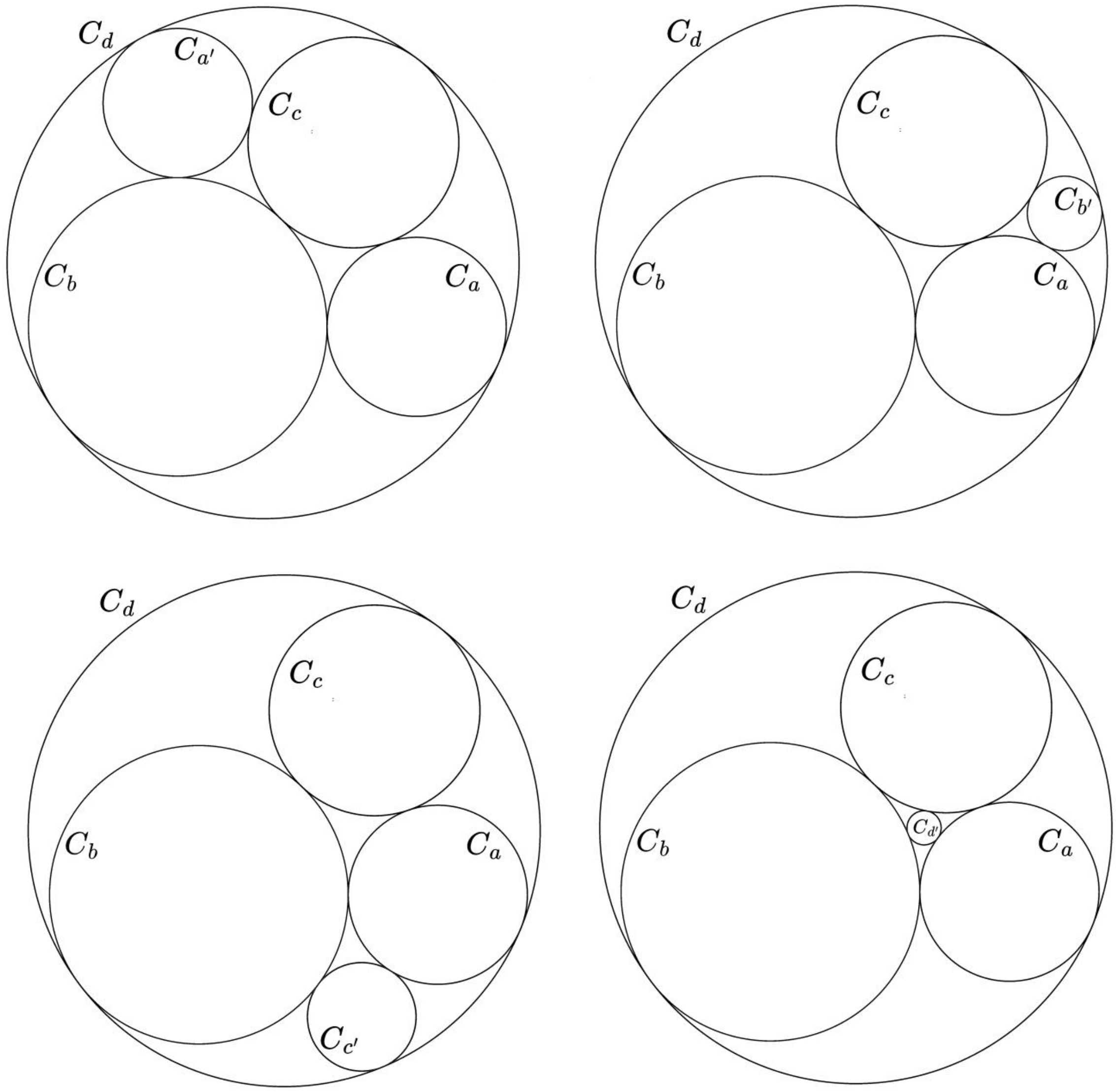

Figure 9

Consider the initial 4-tuple (b_a, b_b, b_c, b_d) as a row vector; one may use matrices to describe the effect of S_a, S_b, S_c and S_d with their actions done by matrix multiplication on the right-hand side of the 4-tuple, where

$$S_a = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{pmatrix},\ S_b = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{pmatrix},$$

$$S_c = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 1 \end{pmatrix},\ S_d = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Alternatively, one may take the initial 4-tuple as a column vector, take the transpose of the above matrices, and multiply on the left-hand side. But the choice of using row vectors here actually follows convention as in [3].

Since $S_a^2 = S_b^2 = S_c^2 = S_d^2 = I$ one shall note that actions of S_a, S_b, S_c and S_d on 4-tuples are also involutory. Note also that entries of these matrices are all integers. Hence, if the initial 4-tuple of curvatures consists only of integers, then the resultant 4-tuple of curvatures under these actions, as well as any of their compositions, will also consist only of integers.

In addition to 4-tuple of curvatures, the center position of a companion circle can be computed in a similar manner following an extension of the Descartes' Circle Theorem in [4]. Combining all these information, one can construct a collection of circles generated with S_a, S_b, S_c and S_d. When the generation process continues, one obtains a circle packing, named after Apollonius of Perga to honor his early work on touching objects, the **Apollonian circle packing** (or **Apollonian gasket**). And all compositions of actions form a *group* called the **Apollonian group**.

Activity 10. Following Activity 9, an integral Apollonian packing is an Apollonian circle packing with curvatures of circles all being integers. With the code shared in the repository, try to illustrate some integral Apollonian Packing.

Check and Discussion 10. Some integral Apollonian packings at different scales are captured in a gallery in Figure 10. The key is to try out some integral values of 4-tuple of curvatures to start from, the initial 4-tuples in the gallery are (3, 6, 7, -1), (12, 20, 17, -7) and (2, 2, 3, -1).

I hope readers enjoy this journey, a first look on integral Apollonian packings. Note also that there is an alternative method for the key calculation in checking Descartes' Circle Theorem using Heron's formula. Interested readers may refer to [5].

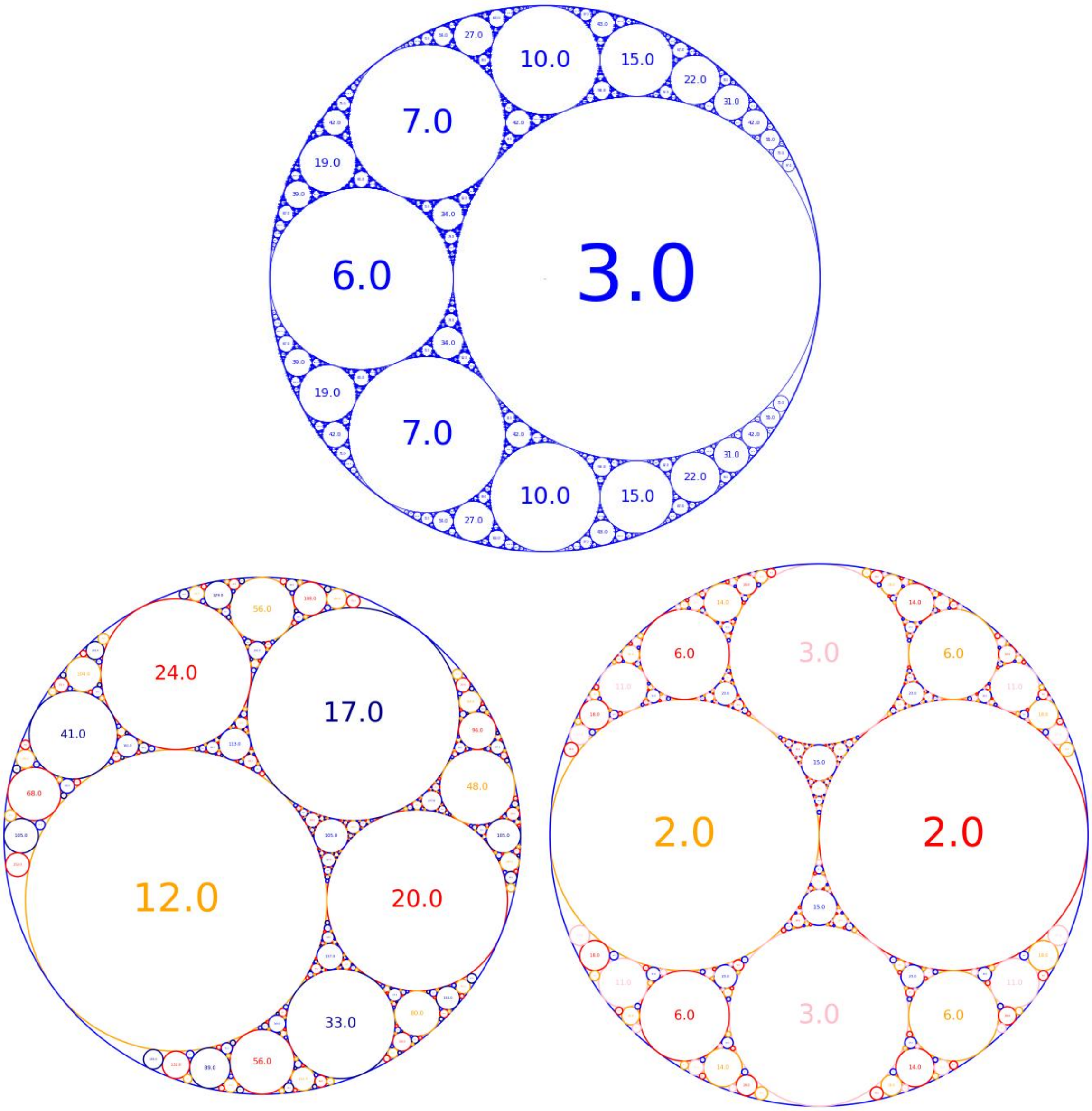

Figure 10

Desserts: Two More Activities Connecting Mathematics and Programming

Now that we have finished the main dish, let's have some desserts. The following two activities come from two other topics the author prepared even earlier. At that time, a chatbot called Sage served as the coding copilot. All in all, the experience of using AI in coding helps one quickly become familiar with syntax in different packages and write sample setups for plotting. Let’s be brief.

Activity 11. Go through the shared notebook *Information_Retrieval.ipynb*. Discuss with your peers how polynomials could be used to design a robust storage scheme.

Activity 12. Mathematics is beautiful and programming is powerful. Please go through *From_Spirographs_to_a_Programmable_Environment.ipynb*, another shared notebook. Use the code there to plot some parametrized curves, including the cardioid, epitrochoid and some curves constructed with more rolling wheels.

Conclusion

In the past, people usually use computers to create their own presentations, from graphical displays to animations. This article aims for a more ambitious approach to make fuller use of computers, on simulations and in a programmable environment. In such an environment, one designs activities to overcome challenges and bypass obstacles. Those interconnections in activities showcase some beautiful minds that one can cultivate, just like an invariant inheritance from the development of mathematics. While it is well known that contemporary generative AIs are built on mathematics; here AIs are in turn applied to work behind, on helping to bridge the gap between mathematical ideas and programming code. For the readers, going through this innovative journey of the interplay is also proof of perseverance. With further community engagement, one hopes the resultant set will be a synergy of technological advances, new productive forces, and talent education, much like a symphony.

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